CLASS OUTLINES FOR HONORS CALCULUS III.

What these are for. The outlines are to orient yourselves about what we have done and where the course is going, especially in case you have to miss a lecture. They may also help in preparing for exams: if something is mentioned here, you definitely have to learn it, together with its proof, which has been explained in class and can be found in the textbook following the references. On the other hand, it is possible that I have mentioned something in class and forgot to put it here, in which case it is still examinable. If you notice that something substantial is missing, feel free to let me know. Everything from the homework problems is also examinable, unless it bears the disclaimer "optional".

This document is intended to help you do well in the course, not to be a literal transcription of everything that happened in lectures. Please take it with a grain of salt. More specifically, I will try to be accurate in what follows, but it is possible there will be typos or mistakes.

Lecture 1, 03/20/2023. Definition of what it means for a sequence of functions $f_n : X \to \mathbf{R}$ to converge pointwise to a function $f : X \to \mathbf{R}$.

Pointwise convergence need not preserve good properties of functions. Assume that $f_n \to f$ pointwise on X. Then:

(1) if f_n is continuous for all n, then f need not be continuous. Example: $f_n(x) = x^n$ on $X = [0,1]$ converges to a function with a jump discontinuity at $x = 1$.

(2) if f_n is differentiable for all n, then f need not be differentiable. The previous example also shows this, and otherwise you can consider

$$\begin{aligned} f_n(x) &= 1 \text{ if } x \in [1/n, 1] \\ &= \sin\frac{n\pi x}{2} \text{ if } x \in [-1/n, 1/n] \\ &= -1 \text{ if } x \in [-1, -1/n]. \end{aligned}$$

Even if f is differentiable, it need not be the case that $f_n' \to f'$. For example, consider

$$f_n(x) = \frac{1}{n}\sin(n^2 x)$$

for $x \in \mathbf{R}$. This converges pointwise to zero (in fact, it does so uniformly), but the sequence of derivatives is

$$f_n'(x) = n\cos(n^2 x)$$

which does not converge to zero: for example, $f_n'(0) \to +\infty$ as $n \to +\infty$.

(3) if $X = [a,b]$ and f_n is integrable on X for all n, then f need not be integrable on $[a,b]$. For example, if $x_0, x_1, x_2, \ldots$ is an enumeration of the rational numbers in $[0,1]$, then

$$f_n(x) = \chi_{x_0}(x) + \cdots + \chi_{x_n}(x)$$

is a sequence of integrable functions on $[0,1]$ converging pointwise to the Dirichlet function $\chi_{\mathbf{Q}}$, which is not integrable.

Even if f is integrable it need not be the case that $\lim_{n\to+\infty} \int_a^b f_n = \int_a^b f$. For example, consider

$$\begin{aligned} f_n(x) &= n(n+1) \text{ if } x \in [1/(n+1), 1/n] \\ &= 0 \text{ if } x \notin [1/n+1, 1/n]. \end{aligned}$$

Then f_n converges to zero pointwise, but

$$\int_0^1 f_n(t)dt = n(n+1)(1/n - 1/(n+1)) = 1$$

and so $\lim_{n\to+\infty} \int_0^1 f_n = 1 \neq \int_0^1 f$.

We ended this lecture with the definition of what it means for f_n to converge uniformly to f on X.

Lecture 2, 03/22/2023. This lecture was about three theorems showing how uniform convergence fixes the issues with pointwise convergence from Lecture 1.

Theorem: Assume that $f_n \to f$ uniformly on X and f_n is continuous for all n. Then f is continuous on X. (Chapter 24, Theorem 2. The proof uses an "$\epsilon/3$-argument", i.e. it uses the triangle inequality twice.)

Theorem: Let f_n and f be integrable functions on $[a, b]$. Assume that $f_n \to f$ uniformly on X. Then

$$\lim_{n\to+\infty} \int_a^b f_n(t)dt = \int_a^b f(t)dt.$$

(Chapter 24, Theorem 1.)

The situation for derivatives is more complicated: uniform convergence of f_n does not imply that $f_n' \to f'$. For example, $f_n = \sin(n^2 x)/n$ converges uniformly to zero on $\mathbf{R}$, but the sequence f_n' of derivatives does not even converge pointwise. The situation is better if we take the convergence of f_n' as an *assumption*, as in the following theorem.

Theorem: Let $f_n : [a, b] \to \mathbf{R}$ be a sequence of differentiable functions converging pointwise to f. Assume that f_n' is integrable for all n, and that f_n' converges uniformly to a continuous function g. Then f is differentiable, and $g = f'$: it follows that

$$f'(x) = \lim_{n\to\infty} f_n'(x) \text{ for all } x \in [a, b].$$

(Chapter 24, Theorem 3. You get the same conclusion by assuming that f_n' is continuous for all n and f_n' converges uniformly to some g: indeed, it then follows that g is continuous by the first theorem in this lecture, and so you don't have to check it beforehand.)

Lecture 3, 03/24/2023. This lecture was about examples, as well as methods for proving uniform convergence.

Theorem: Let $f_n : X \to \mathbf{R}$ be a sequence of functions converging pointwise to f. Then the convergence is uniform if and only if

$$\lim_{n\to\infty} \sup_{x\in X} |f_n(x) - f(x)| = 0.$$

This is essentially a restatement of various definitions, and I encourage you to write your own proof before reading what follows: it will be a good test of your understanding of the definitions.

Proof. Assume uniform convergence. Fix $\epsilon > 0$. Since $f_n \to f$ uniformly, there exists $N > 0$ such that

$$n \geq N \text{ implies } |f_n(x) - f(x)| < \epsilon \text{ for all } x \in X.$$

Hence if $n \geq N$ then

$$\sup_{x\in X} |f_n(x) - f(x)| \leq \epsilon.$$

It follows that $\sup_{x\in X} |f_n(x)-f(x)| \to 0$ as $n \geq N$, by definition of what it means for a sequence to converge to zero.

Now assume that $\lim_{n\to\infty} \sup_{x\in X} |f_n(x) - f(x)| = 0$. Fix $\epsilon > 0$. Then there exists $N > 0$ such that if $n \geq N$ then

$$\sup_{x\in X} |f_n(x) - f(x)| < \epsilon.$$

Hence for all $z \in X$ we have that $n \geq N$ implies

$$|f_n(z) - f(z)| < \epsilon.$$

and this is the definition of uniform convergence. □

The following examples will use the theorem above.

1. If $\sup_{x\in X} |f_n(x)| \to 0$ as $n \to +\infty$ then $f_n \to 0$ uniformly on X. Example: $f_n(x) = \sin(n^2 x)/n$ and $g_n(x) = e^{-x^2}/n$ converge uniformly to zero on $\mathbf{R}$.
2. Typically, Taylor series do not converge uniformly on $\mathbf{R}$. For example, let

$$f_n(x) = 1 + x + \cdots + \frac{x^n}{n!}$$

so that $f_n \to \exp$ pointwise on $\mathbf{R}$. By the Lagrange form of the remainder, if $x > 0$ we have

$$\exp(x) - f_n(x) = \exp(t_n)\frac{x^{n+1}}{(n+1)!}$$

for some $t_n \in (0, x)$. Since $t_n > 0$ we have $\exp(t_n) > 1$, and so

$$|\exp(x) - f_n(x)| \geq \frac{x^{n+1}}{(n+1)!} \text{ for all } x > 0.$$

Hence

$$\sup_{x \in \mathbf{R}} |\exp(x) - f_n(x)| \to +\infty$$

as $n \to +\infty$, and the convergence is not uniform.

We will see next time that power series converge uniformly on *closed and bounded* intervals.

(3) The sequence

$$f_n(x) = \sqrt{x^2 + \frac{1}{n^2}}$$

converges uniformly to $f(x) = |x|$ on $[-1, 1]$. (Notice that this is an example of a uniform limit of differentiable functions that is not differentiable.)

To prove this, we use the inequality

$$x^2 \leq x^2 + \frac{1}{n^2} \leq x^2 + \frac{2|x|}{n} + \frac{1}{n^2}$$

which implies

$$|x| \leq f_n(x) \leq |x| + \frac{1}{n}$$

and so

$$|f_n(x) - |x|| \leq \frac{1}{n}$$

for all $x \in [-1, 1]$. Hence

$$\sup_{x \in [-1,1]} |f_n(x) - |x|| \leq 1/n \to 0 \text{ as } n \to +\infty.$$

(4) The mean value theorem can help in establishing uniform convergence via the theorem above. For example, consider

$$f_n(x) = \log(x + \frac{1}{n^2}) \text{ for } x > 1.$$

Then $f_n \to \log$ pointwise. The mean value theorem implies

$$|f_n(x) - \log(x)| = |\log(x + \frac{1}{n^2}) - \log(x)| = \frac{1}{t_n n^2}$$

for some $t_n \in (x, x + 1/n^2)$. Since $x > 1$, this implies $t_n^{-1} < 1$, and so

$$\sup_{x>1} |f_n(x) - \log(x)| < \frac{1}{n^2} \to 0 \text{ as } n \to +\infty.$$

We conclude by defining what it means for a series of functions to converge pointwise and uniformly, and stating analogues of the theorems of Lecture 2 for series of functions. (See Chapter 24, Corollary for this.)

Lecture 4, 03/27/2023. We started this lecture with the proof of the following theorem, which provides a very useful method to check uniform convergence.

Theorem (Weierstrass M-test): Let $\sum_{n \geq 0} f_n$ be a series of functions on X. Assume that there exists a sequence of numbers M_n such that

(1) $|f_n(x)| \leq M_n$ for all $x \in X$, and
(2) the series $\sum_{n \geq 0} M_n$ converges. (Notice that the previous part implies $M_n \geq 0$ for all n, hence this is the same as absolute convergence.)

Then $\sum_{n\geq 0} f_n$ converges absolutely and uniformly on X. (Chapter 24, Theorem 4. Make sure not to confound absolute convergence and uniform convergence: they are different concepts and behave in different ways.)

Optional application: There exists a continuous function $f : \mathbf{R} \to \mathbf{R}$ which is not differentiable at any point. See Chapter 24, Theorem 5 for a constructive proof.

We then moved to the study of power series, which are a particularly important example of series of functions.

Definition: A real power series is a series of functions of the form $\sum_{n\geq 0} \alpha_n (x-t)^n$ for some $t, \alpha_n \in \mathbf{R}$. We will typically work with $t = 0$.

Example: Let $f : (a, b) \to \mathbf{R}$ be an infinitely differentiable function, and let $t \in (a, b)$. Then the Taylor series of f at t is defined to be the power series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(t)}{n!}(x-t)^n.$$

When making this definition we are not claiming anything about the convergence of the Taylor series, and we are not asserting that it converges to the original function f. In fact, this need not be true: the function $f(x) = e^{-1/x^2}$ gives a counterexample at zero, as seen in Math 16200.

Theorem (Chapter 24, Theorem 6): Assume that $x_0 \in \mathbf{R}$ and the series $\sum_{n\geq 0} \alpha_n x_0^n$ converges (absolutely or not). Assume that $0 < z < |x_0|$. Then the power series $\sum_{n\geq 0} \alpha_n x^n$ converges absolutely and uniformly on $[-z, z]$.

Remark: This theorem does not say that the power series converges on $[-x_0, x_0]$. In fact, a counterexample is provided by the Taylor series for $\log(1 + x)$ at 0, which is

$$\sum_{n\geq 1} (-1)^{n+1} \frac{x^n}{n}.$$

In fact, this power series converges at $x_0 = 1$ (by Leibniz's theorem) but not at $x_0 = -1$ (since the harmonic series diverges).

Lecture 5, 03/29/2023. We introduced the notion of radius of convergence of a power series.

Theorem: Let $F(x) = \sum_{n\geq 0} \alpha_n x^n$ be a power series. Then there exists a unique $R \geq 0$, called the radius of convergence of F and possibly equal to $+\infty$, such that

$$|x| < R \text{ implies that } F(x) \text{ converges}$$
$$|x| > R \text{ implies that } F(x) \text{ diverges.}$$

In fact, if $|x| < R$ then $F(x)$ converges absolutely and uniformly on $[-x, x]$.

Proof. Uniqueness: assume that R_1 and R_2 have the two properties in the statement of the theorem. If $R_1 \neq R_2$ then without loss of generality we can assume that $R_1 < R_2$. Let $x \in (R_1, R_2)$. Since $|x| > R_1$, the second property for R_1 implies that $F(x)$ diverges. Since $|x| < R_2$, the first property for R_2 implies that $F(x)$ converges. So $F(x)$ converges and diverges at the same time, which is a contradiction, so $R_1 = R_2$.

Existence: let

$$R = \sup\{|x| : F(x) \text{ converges}\}.$$

If $|x| > R$ then $F(x)$ diverges by definition of the supremum of a set of real numbers. Assume now that $|x| < R$. Then by definition of the supremum there exists $\rho \in (|x|, R)$ such that $F(\rho)$ converges. By a theorem in the previous lecture, it follows that the power series F converges absolutely and uniformly on $[-x, x]$. □

Corollary: Let $\sum_{n\geq 0} \alpha_n x^n$ be a power series with radius of convergence R. Then the formula

$$f(x) = \sum_{n\geq 0} \alpha_n x^n$$

defines a continuous function $f : (-R, R) \to \mathbf{R}$.

Proof. If $|x| < R$ then $\sum_{n\geq 0} \alpha_n x^n$ converges at x by definition of the radius of convergence, and so f is a well-defined function. To prove continuity, it is equivalent to prove that if $x_n \in (-R, R)$ is a sequence with limit $x \in (-R, R)$ then $\lim_{n\to+\infty} f(x_n) = f(x)$. Since $x \in (-R, R)$, there exists $\delta > 0$ such that $x \in$

$[-R+\delta, R-\delta]$. (Draw a picture if this isn't clear: the point is that x is contained in a slightly smaller, closed and bounded interval contained in $(-R, R)$.)

Since $x_n \to x$, there exists $N > 0$ such that $n \geq N$ implies $x_n \in [-R+\delta, R-\delta]$. Now we use the fact that $|R-\delta| < R$, and so by the previous theorem the power series $\sum_{n\geq 0} \alpha_n x^n$ converges uniformly to f on $[-R+\delta, R-\delta]$. Hence the limit function f is continuous on $[-R+\delta, R-\delta]$, and so $\lim_{n\to\infty} f(x_n) = f(x)$. $\square$

Remark: HW2 contains a useful formula to compute the radius of convergence of a power series using the ratio test.

Our next theorem shows that much more is true, and in fact a power series defines an infinitely differentiable function inside its own interval of convergence. The main idea is to try to differentiate the the function by differentiating the power series termwise. More precisely, if $F(x) = \sum_{n\geq 0} \alpha_n x^n$ is a power series, we define the formal derivative (or derived series) of F by

$$F'(x) = \sum_{n\geq 1} n\alpha_n x^{n-1}.$$

Theorem: Let R (resp. R') be the radius of convergence of F (resp. F'). Then $R \leq R'$.

Proof. Since both $R, R' \geq 0$, to prove the inequality $R \leq R'$ it suffices to prove the implication

$$\text{for all } x \geq 0, x < R \Rightarrow x \leq R'.$$

In fact, if this is true then it follows that

$$R = \sup\{x \geq 0 : x < R\} \leq \sup\{x \geq 0 : x \leq R'\} = R'.$$

So assume that $x \geq 0$ and $x < R$. Choose a number $\rho \in (x, R)$. Then the series $\sum_{n\geq 0} \alpha_n \rho^n$ converges, and so its terms are bounded: there exists $M > 0$ such that

$$|\alpha_n \rho^n| < M$$

for all n. It follows that

$$|n\alpha_n x^{n-1}| = n|\alpha_n||x|^{n-1} = \frac{1}{|x|} n|\alpha_n||\rho^n||x/\rho|^n \leq \frac{M}{|x|} n|x/\rho|^n.$$

Our assumption that $x < \rho$ implies that the series $\sum_{n\geq 0} n|x/\rho|^n$ converges by the ratio test: in fact,

$$\lim_{n\to\infty} \frac{n+1}{n} |x/\rho| = |x/\rho| < 1.$$

It follows that

$$F'(x) = \sum_{n\geq 0} n\alpha_n x^{n-1}$$

converges absolutely by the comparison test. By definition of R', this implies that $x \leq R'$, which is what we had to prove. $\square$

Remark (optional): in fact we always have $R = R'$. This can be proved introducing the formal integral

$$\int F = \sum_{n\geq 0} \alpha_n \frac{x^{n+1}}{n+1}$$

and proving by the same method that the radius of convergence of $\int F$ is at least R. In fact, this will implies that

$$\text{radius}(\int F') \geq \text{radius}(F') \geq \text{radius}(F)$$

but $\int F' = F - \alpha_0$, and so radius$(\int F') = $ radius(F). So we get $R \geq R' \geq R$ and so $R = R'$.

Corollary: The function

$$f(x) = \sum_{n\geq 0} \alpha_n x^n$$

is differentiable in the open interval $(-R, R)$, with derivative given by the formula

$$f'(x) = \sum_{n\geq 1} \alpha_n n x^{n-1}.$$

Proof. Since $R' \geq R$, the power series F' converges absolutely and uniformly on $(-R+\epsilon, R-\epsilon)$ for all $\epsilon > 0$ small enough, and the limit function is continuous. By the theorem on uniform convergence of derivatives, it follows that f is differentiable on $(-R, R)$ and its derivative at $x \in (-R, R)$ is given by the sum of the series $F'(x)$. □

Iterating this argument (or using induction) we get the following corollary.

Corollary: The function f of the previous corollary is infinitely many times differentiable on $(-R, R)$, and its n-th derivative is given by the sum of the series $F^{(n)}$ obtained by differentiating F termwise n times.

This corollary allows you to recover the coefficients of F from the function f it defines, provided the radius of convergence of F is not zero. More precisely, we get

$$f^{(n)}(0) = F^{(n)}(0) = n(n-1)\cdots 1\alpha_n$$

and so

$$\alpha_n = \frac{f^{(n)}(0)}{n!}.$$

It follows that if a function f is given by a convergent power series F, then F must be the Taylor series of f. The following consequence if useful in applications: if $F = \sum_{n\geq 0} \alpha_n x^n$ and $G = \sum_{n\geq 0} \beta_n x^n$ are two power series with positive radius of convergence, and the functions they define coincide on an open interval containing 0, then $F = G$. Indeed, we find that

$$\alpha_n = \frac{f^{(n)}(0)}{n!} = \frac{g^{(n)}(0)}{n!} = \beta_n$$

for all n.

Lecture 6, 03/31/2023. This lecture was about the Euclidean space $\mathbf{R}^n$ together with its operations of vector sum and scalar multiplication, and inner product and Euclidean norm. These are explained in the section "Norm and inner product" of the book *Calculus on manifolds*, and I will not repeat them here. An important result is the following theorem, which is called the Cauchy–Schwarz inequality in n dimensions. It allows us to define the angle θ between two vectors x, y not contained in the same line, by the formula

$$\theta = \arccos^{-1}(\langle x, y\rangle / ||x||||y||).$$

Theorem: Let $x, y \in \mathbf{R}^n$ be nonzero vectors. Then

$$|\langle x, y\rangle| \leq ||x||||y||,$$

with equality if and only if there exists $\lambda \in \mathbf{R}$ such that $y = \lambda x$.

Proof. Assume first that $y = \lambda x$ for some $\lambda \in \mathbf{R}$. Then

$$\langle x, y\rangle = \langle x, \lambda x\rangle = \lambda ||x||^2.$$

But $||x||||y|| = |\lambda|||x||^2$, and since $||x||^2 \geq 0$ we deduce $|\langle x, y\rangle| = ||x||||y||$.

On the other hand, if the equation $y = \lambda x$ has no solutions $\lambda \in \mathbf{R}$ then $||y - \lambda x|| \neq 0$ for all $\lambda \in \mathbf{R}$. Squaring and expanding out we find that

$$(y - \lambda x) \cdot (y - \lambda x) = y \cdot y - 2\lambda x \cdot y + \lambda^2 x \cdot x \neq 0$$

for all $\lambda \in \mathbf{R}$. Hence the quadratic equation in λ given by

$$||x||^2\lambda^2 - 2\langle x, y\rangle\lambda + ||y||^2 = 0$$

has no solutions $\lambda \in \mathbf{R}$. Hence its discriminant is strictly negative, which means that

$$4\langle x, y\rangle^2 < 4||x||^2||y||^2.$$

Dividing by 4 and taking square roots yields the theorem. □

The properties of $\mathbf{R}^n$ do not include a vector multiplication, and we have seen that the obvious candidate (coordinatewise multiplication) does not have the property that nonzero vectors are invertible. A special case occurs when $n = 2$: define

$$(x, y)(\alpha, \beta) = (x\alpha - y\beta, x\beta + y\alpha).$$

The Euclidean space $\mathbf{R}^2$ equipped with all the previous operations together with this multiplication is called the field of complex numbers and denoted $\mathbf{C}$. We have checked that $i = (0,1)$ squares to -1, and that if we write (x,y) as $x+iy$ then these behave like the complex numbers you are used to.

Lecture 7, 04/03/2023. This lecture was about functions on $\mathbf{R}^n$. Every function $\mathbf{R}^n \to \mathbf{R}^m$ is built out of functions $\mathbf{R}^n \to \mathbf{R}$ in the following way.

Lemma: Let $f : \mathbf{R}^n \to \mathbf{R}^m$ be a function. Then there exist functions

$$f_1, f_2, \dots, f_m : \mathbf{R}^n \to \mathbf{R}$$

such that $f(x) = (f_1(x), f_2(x), \dots, f_m(x))$ for all $x \in \mathbf{R}^n$.

Proof. If $1 \le i \le m$, define the i-th coordinate function

$$c_i = \mathbf{R}^m \to \mathbf{R}, c_i(x_1, \dots, x_m) = x_i.$$

If we put $f_i = c_i \circ f$, then it follows from the definitions that $f(x) = (f_1(x), \dots, f_m(x))$ for all $x \in \mathbf{R}^n$. □

Definition: A function $\mathbf{R}^n \to \mathbf{R}^m$ is linear (more precisely, $\mathbf{R}$-linear) if for all $x, y \in \mathbf{R}^n$ and $\lambda \in \mathbf{R}$ we have

$$f(x+y) = f(x) + f(y) \text{ and } f(\lambda x) = \lambda f(x).$$

Definition: Let $1 \le i \le n$. The vector $e_i \in \mathbf{R}^n$ has all coordinates equal to zero except for the i-th coordinate, which is equal to 1. The set $\{e_1, \dots, e_n\}$ is called the standard basis of $\mathbf{R}^n$.

Theorem: Let $v_1, v_2, \dots, v_n \in \mathbf{R}^m$. Then there exists a unique linear map

$$f : \mathbf{R}^n \to \mathbf{R}^m$$

such that $f(e_i) = v_i$ for all $1 \le i \le n$.

Proof. The proof will use the following fact, which follows from the definitions: If $x = (x_1, \dots, x_n) \in \mathbf{R}^n$, then $x = x_1 e_1 + x_2 e_2 + \dots + x_n e_n$.

Uniqueness: We will prove that if $f, g : \mathbf{R}^n \to \mathbf{R}^m$ are linear maps with $f(e_i) = g(e_i)$ for all $1 \le i \le n$ then $f = g$. Let $x \in \mathbf{R}^n$. Then

$$f(x) = f(x_1 e_1 + \dots + x_n e_n) = \sum_i x_i f(e_i) = \sum_i x_i g(e_i) = g(x_1 e_1 + \dots + x_n e_n) = g(x).$$

Hence $f(x) = g(x)$ for all $x \in \mathbf{R}^n$, and so $f = g$.

Existence: If $x \in \mathbf{R}^n$, define

$$f(x) = x_1 v_1 + x_2 v_2 + \dots + x_n v_n.$$

Then $f(e_i) = v_i$ by definition. There remains to check that f is linear, which is left for you as an exercise. □

Lecture 8, 04/05/2023. It follows from the previous lecture that if $f : \mathbf{R}^n \to \mathbf{R}^m$ is linear then all the information about f is contained in the vectors $f(e_1), f(e_2), \dots, f(e_n)$. We arrange these vectors as columns of a matrix, making the following definition.

Definition: Let $f : \mathbf{R}^n \to \mathbf{R}^m$ be a linear map. The matrix $[f] \in M_{m \times n}(\mathbf{R})$ attached to f is the matrix whose i-th column is $f(e_i)$.

Definition: Let $A \in M_{m \times n}(\mathbf{R})$ and let $v \in \mathbf{R}^n$. Write their coordinates as $A = (\alpha_{ij})$ and $v = (v_1, \dots, v_n)$. Then we define the product Av by the formula

$$(Av)_i = \sum_{j=1}^n \alpha_{ij} v_j.$$

In other words, the i-th coordinate of Av is the dot product of the i-th row of A with v: this is the matrix-vector product you might already be used to.

Theorem: 1) Let $A \in M_{m \times n}(\mathbf{R})$. Then the formula $f_A(v) = Av$ defines a linear map $f_A : \mathbf{R}^n \to \mathbf{R}^m$, and $[f_A] = A$.

2) Let $g : \mathbf{R}^n \to \mathbf{R}^m$ be linear. Then $g(v) = [g]v$ for all $v \in \mathbf{R}^n$, and so $g = f_{[g]}$.

Proof. Both parts are direct calculations using the following alternative interpretation of the matrix-vector product: if $v = (v_1, \ldots, v_n)$ and we write A_i for the i-th column of $A \in M_{m\times n}(\mathbf{R})$, then you can check that

$$Av = v_1 A_1 + v_2 A_2 + \cdots + v_n A_n.$$

Proof of (1): Fix $v, w \in \mathbf{R}^n$ and $\lambda \in \mathbf{R}$. To check that f_A preserves vector sum we compute

$$f_A(v+w) = A(v+w) = \sum_{i=1}^{n} (v+w)_i A_i = \sum_{i=1}^{n} (v_i A_i + w_i A_i) = Av + Aw.$$

You should check that $f_A(\lambda v) = \lambda f_A(v)$ as an exercise. These two properties imply that f_A is linear. Finally, by definition $[f_A]$ is the matrix whose i-th column is $f_A(e_i)$, so we need to check that $f_A(e_i) = A_i$. This is true because

$$f_A(e_i) = \sum_{j=1}^{n} (e_i)_j A_j$$

and $(e_i)_j = 0$ if $i \neq j$ and 1 if $i = j$.

Proof of (2): We need to prove that $g = f_{[g]}$. Since both sides are linear maps, by a theorem from last time it suffices to check that $g(e_i) = f_{[g]}(e_i)$ for all i. But $f_{[g]}(e_i) = [g]e_i$ by definition, and by the same computation as in part (1) we know that $[g]e_i$ is the i-th column of $[g]$. By definition this is $g(e_i)$, and so we are done. □

Corollary: There is a bijection from the set of linear maps $\mathbf{R}^n \to \mathbf{R}^m$ to $M_{m\times n}(\mathbf{R})$, sending f to $[f]$.

Proof. By the previous theorem, the inverse bijection sends A to f_A. □

Lecture 9, 04/07/2023. No lecture, first midterm.

Lecture 10, 04/10/2023. We started this lecture with the proof of the theorem from last time, and then introduced matrix multiplication and composition of functions. Matrix multiplication is only defined for pairs of matrices satisfying a certain compatibility condition, as in the following definition.

Definition: If $A = (\alpha_{ij}) \in M_{m\times n}(\mathbf{R})$ and $B = (\beta_{ij}) \in M_{n\times t}(\mathbf{R})$ then the matrix product $AB = (\gamma_{ij})$ is defined by

$$\gamma_{ij} = \sum_{k=1}^{n} \alpha_{ik}\beta_{kj}.$$

Equivalently: the ij-th entry of AB is the dot product of the i-th row of A with the j-th column of B. Equivalently: AB has as many columns as B, and the j-th column of AB is A(j-th column of B).

Theorem: Let $f : \mathbf{R}^t \to \mathbf{R}^n$ and $g : \mathbf{R}^n \to \mathbf{R}^m$ be linear maps. Then the composition $g \circ f$ is linear, and $[g \circ f] = [g][f]$.

Proof. Let $v, w \in \mathbf{R}^t$ and $\lambda \in \mathbf{R}$. Then

$$(g \circ f)(v+w) = g(f(v+w)) = g(f(v) + f(w)) = g(f(v)) + g(f(w)) = (g\circ f)(v) + (g \circ f)(w).$$

You should check that $(g \circ f)(\lambda v) = \lambda(g \circ f)(v)$ as an exercise. These properties imply that $g \circ f$ is linear.

To prove that $[g \circ f] = [g][f]$ it suffices to see that they have the same columns, and so it suffices to prove that

$$[g \circ f]e_i = ([g][f])e_i$$

for all e_i. If $v \in \mathbf{R}^n$ then we know that $[g]v = g(v)$ by a theorem from the previous lecture. Letting $v = [f](e_i)$ we find that

$$([g][f])e_i = [g]([f]e_i) = g([f]e_i) = g(f(e_i))$$

since $[f]e_i = f(e_i)$ by definition. We conclude by noticing that $g(f(e_i)) = [g \circ f](e_i)$ by definition of $[g\circ f]$. □

We finished the lecture with an introduction to continuity in n-dimensions.

Lecture 11, 04/12/2023. Definition: Let $X \subset \mathbf{R}^n$ and let $f : X \to \mathbf{R}^m$ be a function. Choose $x \in X$. We say that f is continuous at $x \in X$ if for all $\epsilon > 0$ there exists $\delta > 0$ such that if $y \in X$ and $||y - x|| < \delta$ then $||f(y) - f(x)|| < \epsilon$.

Theorem: Let $f : X \to \mathbf{R}^m$ be a function. Then f is continuous if and only if $f_i = c_i \circ f$ for all $1 \leq i \leq m$, where c_i is the i-th coordinate function on $\mathbf{R}^m$.

Theorem: Let $f : \mathbf{R}^n \to \mathbf{R}^m$ be a linear map. Then f is continuous.

Both theorems rely upon a geometric property of distance: If $x \in \mathbf{R}^n$, then

$$\max\{|x_1|, |x_2|, \dots, |x_n|\} \leq ||x|| \leq \sqrt{n} \max\{|x_1|, \dots, |x_n|\}.$$

Proof. Let i be such that

$$|x_i| = \max\{|x_1|, |x_2|, \dots, |x_n|\}.$$

Since $x_i^2 \leq x_1^2 + \cdots + x_n^2$ we find that $|x_i| \leq ||x||$ by taking square roots. Since by construction $|x_j| \leq |x_i|$ for all j, we have

$$x_1^2 + \cdots + x_n^2 \leq n x_i^2$$

and so $||x|| \leq \sqrt{n}|x_i|$ follows by taking square roots. □

Proof of the first theorem. Fix $\epsilon > 0$ and let $x \in X$. Assume that f is continuous. We are going to prove that f_i is continuous at x for all i. Since x is arbitrary, this will imply the forward direction. By assumption, there exists $\delta > 0$ such that $||y - x|| < \delta$ and $y \in X$ implies $||f(y) - f(x)|| < \epsilon$. By the first inequality above, we find that $\max_i\{|f(y)_i - f(x)_i|\} < \epsilon$ if $y \in X, ||y - x|| < \delta$. Hence $y \in X, ||y - x|| < \delta$ implies $|f_i(y) - f_i(x)| < \epsilon$ for all i. This means that f_i is continuous at x for all i.

Now assume that f_i is continuous for all i. We will prove that f is continuous at x. By assumption, for all i there exists $\delta_i > 0$ such that $y \in X, ||y - x|| < \delta_i$ implies $|f(y)_i - f(x)_i| < \epsilon/\sqrt{n}$. Let $\delta = \min\{\delta_1, \dots, \delta_n\}$. By the second inequality above, we find that $y \in X, ||y - x|| < \delta$ implies

$$||f(y) - f(x)|| \leq \sqrt{n} \max_i\{|f_i(y) - f_i(x)|\} < \sqrt{n}\epsilon/\sqrt{n} = \epsilon.$$

This means that f is continuous at x. □

Remark: Using this theorem it is easy to give a proof of the continuity of linear functions: their coordinates are polynomial functions (in fact, they have degree one and no constant term) and so by a HW5 exercise they are continuous. We will give a different proof below, which has the advantage of proving that linear maps are in fact uniformly continuous.

Proof of the second theorem. We will prove that there exists $M > 0$ such that $||f(x)|| \leq M||x||$ for all $x \in \mathbf{R}^n$. (The important point here is that M works for all x simultaneously: it is very easy to find such a bound M if you allow it to depend on x.)

Using this fact, we can actually prove that f is uniformly continuous: fix $\epsilon > 0$. Then $||y - x|| < \epsilon/M$ implies

$$||f(y) - f(x)|| = ||f(y - x)|| \leq M||y - x|| < \epsilon$$

and so f is uniformly continuous.

To construct M, it will be easier to work with

$$||x||_\infty = \max\{|x_1|, |x_2|, \dots, |x_n|\}$$

instead of $||x||$, and then translate back to $||x||$ using the geometric properties of distance we proved before. More precisely, we will prove that there exists M' such that

$$||f(x)||_\infty \leq M'||x||_\infty$$

for all $x \in \mathbf{R}^n$. This implies that

$$||f(x)|| \leq \sqrt{m}||f(x)||_\infty \leq M'\sqrt{m}||x||_\infty \leq M'\sqrt{m}||x||$$

and so we can put $M = \sqrt{m}M'$.

To find M' we will work with the matrix $[f]$ of f. Let $M' = n \max_{i,j}\{|\alpha_{ij}|\}$. We need to prove that $||f(x)||_\infty \leq M'||x||_\infty$, or equivalently

$$|f(x)_i| \leq M'||x||_\infty$$

for all values of i. By definition of $[f]$ we have

$$|f(x)_i| = |\alpha_{i1}x_1+\cdots+\alpha_{in}x_n| \leq \sum_{j=1}^{n} |\alpha_{ij}||x_j| \leq \max_{i,j}\{|\alpha_{ij}|\}(|x_1|+\cdots+|x_n|) \leq \max_{i,j}\{|\alpha_{ij}|\}\cdot n\cdot\max\{|x_1|,\ldots,|x_n|\}$$

and so we are done since the right-hand side of expression above is equal to $M'||x||_\infty$.

$\square$

Lecture 12, 04/14/2023. Definition: A sequence $\{x_i\} \subset \mathbf{R}^n$ converges to a point $x \in \mathbf{R}^n$ if for all $\epsilon > 0$ there exists $N > 0$ such that $i \geq N$ implies $||x_i - x|| < \epsilon$. If this is true we write $x = \lim_{i\to\infty} x_i$ or $x_i \to x$.

Theorem: Let $\{x_i\}$ be a sequence in $\mathbf{R}^n$. Then $x_i \to x$ if and only if $c_j(x_i) \to c_j(x)$ for all $1 \leq j \leq n$. (Here c_j is the j-th coordinate function on $\mathbf{R}^n$.)

Proof. Fix $\epsilon > 0$. If $x_i \to x$ then there exists $N > 0$ such that $i > N$ implies $||x_i - x|| < \epsilon$. This implies $||x_i - x||_\infty < \epsilon$, and so $i > N$ implies $|c_j(x_i) - c_j(x)| < \epsilon$ for all j. This means that $c_j(x_i) \to c_j(x)$ as $i \to \infty$ for all j.

Conversely, if $c_j(x_i) \to c_j(x)$ for all j then there exist N_j such that $i > N_j$ implies $|c_j(x_i) - c_j(x)| < \epsilon/\sqrt{n}$. If $N = \max\{N_1, \ldots, N_n\}$ we deduce that $i > N$ implies $||x_i - x||_\infty < \epsilon/\sqrt{n}$. By a lemma from previous lectures we deduce that $i > N$ implies $||x_i - x|| < \epsilon$, and so $x_i \to x$ as $i \to \infty$. $\square$

Definition: 1) An open rectangle in $\mathbf{R}^n$ is a set of the form

$$(\alpha_1, \beta_1) \times \cdots \times (\alpha_n, \beta_n)$$

for real numbers $\alpha_i < \beta_i \in \mathbf{R}$.

2) If $x \in \mathbf{R}^n$ and $\epsilon > 0$, the open ball of radius ϵ and centre x is defined to be

$$B(x, \epsilon) = \{y \in \mathbf{R}^n : ||y - x|| < \epsilon\}.$$

3) A set $X \subset \mathbf{R}^n$ is called an open set if for all $x \in X$ there exists an open rectangle A such that $x \in A$ and $A \subset X$.

Remark: You have proved in HW4 that X is open if and only if for all $x \in X$ there exists $\epsilon_x > 0$ such that $B(x, \epsilon_x) \subset X$. We will often use this characterization of open sets without mentioning it explicitly.

Examples: 1) By definition, open rectangles are open.

2) By a homework exercise, open balls are open.

3) Assume $n = 1$. The open interval $(\alpha, \beta) \subset \mathbf{R}$ is open: it is both an open rectangle and an open ball (the centre is the midpoint). The intervals $[\alpha, \beta), (\alpha, \beta], [\alpha, \beta]$ are not open.

4) Let $\{U_i\}$ be a collection of open sets in $\mathbf{R}^n$. Then the union $\bigcup_i U_i$ is open.

5) Let $\{U_1, U_2, \ldots, U_n\}$ be a finite collection of open sets in $\mathbf{R}^n$. Then $U_1 \cap U_2 \cap \cdots \cap U_n$ is open.

6) $\mathbf{R}^n$ is open. The empty set is open (the proof of this fact is optional).

We ended this lecture by stating a characterization of continuous functions in terms of open sets, which will be proved as the main theorem of the next lecture.

Lecture 13, 04/17/2023. We started this lecture with a review of images and preimages of sets under functions.

Definition: Let $f : X \to \mathbf{R}^n$ be a function. Let $U \subseteq \mathbf{R}^n$. The image of X under f is defined to be

$$f(X) = \{f(x) : x \in X\} = \{y \in \mathbf{R}^n : \text{ there exists } x \in X \text{ such that } y = f(x)\}.$$

The preimage of U under f is defined to be

$$f^{-1}(U) = \{x \in X : f(x) \in U\}.$$

Examples: 1) Let $f : \mathbf{R}^2 \to \mathbf{R}, f(x, y) = y^2 - x$. Then $f^{-1}\{0\}$ is a parabola.

2) Let $g : \mathbf{R}^2 \to \mathbf{R}, g(x, y) = x^2 + y^2$. Then $g^{-1}\{1\}$ is a circle.

3) Let g be as in the previous example. Let U be the open interval $(-1, 1)$. Then $g^{-1}(U)$ is $B(0, 1)$, the open ball of radius 1 centered at zero.

4) Let $\{U_i\}$ be a collection of subsets of $\mathbf{R}^n$. Then

$$f^{-1}\left(\bigcup_i U_i\right) = \bigcup_i f^{-1}(U_i)$$
$$f^{-1}\left(\bigcap_i U_i\right) = \bigcap_i f^{-1}(U_i).$$

5) Let $U \subset \mathbf{R}^n$. Then

$$f^{-1}\left(\mathbf{R}^n \setminus U\right) = X \setminus f^{-1}(U).$$

(Recall that $X \setminus f^{-1}(U)$ is the complement of $f^{-1}(U)$ in X, i.e. the set of all points of X not contained in $f^{-1}(U)$.

Lemma: Let $X \subset \mathbf{R}^n$ and let $f : X \to \mathbf{R}^m$ be a function. Choose $x \in X$. Then f is continuous at x if and only if the following is true: for all open balls $B(f(x), \epsilon)$, there exists an open ball $B(x, \delta)$ such that

$$B(x, \delta) \cap X \subset f^{-1}\left(B(f(x), \epsilon)\right)$$

Proof. This is a restatement of the definition of continuity. Indeed, the condition in the statement of the lemma occurs if and only if for all $\epsilon > 0$ there exists $\delta > 0$ such that

$$y \in B(x, \delta) \cap X \text{ implies } f(y) \in B(x, \epsilon).$$

But this means that $y \in X, ||y - x|| < \delta$ implies $||f(y) - f(x)|| < \epsilon$, which is the definition of continuity of f at x. $\square$

Theorem: Let $X \subset \mathbf{R}^n$ and let $f : X \to \mathbf{R}^m$ be a function. Then f is continuous if and only if the following is true: for all open sets $U \subset \mathbf{R}^m$ there exists an open set $V \subset \mathbf{R}^n$ such that

$$f^{-1}(U) = V \cap X.$$

Proof. Assume the condition of the theorem is true. Let $x \in X$ and fix $\epsilon > 0$. Let $U = B(f(x), \epsilon)$. Since U is open in $\mathbf{R}^m$ there exists an open set $V \subset \mathbf{R}^n$ such that $f^{-1}(U) = V \cap X$, and since $x \in f^{-1}(U)$ it follows that there exists $\delta > 0$ such that $B(x, \delta) \subset V$. This implies that $B(x, \delta) \cap X \subset f^{-1}(U)$, and so by the lemma above we conclude that f is continuous at x.

Conversely, assume that f is continuous. Let $U \subset \mathbf{R}^m$ be open. If $f^{-1}(U)$ is empty, then we are done since we can let V be the empty set, which is open. Otherwise, choose $x \in f^{-1}(U)$. Since U is open, there exists $\epsilon_x > 0$ such that

$$B(x, \epsilon_x) \subset U.$$

Since f is continuous at x, there exists $\delta_x > 0$ such that

$$B(x, \delta_x) \cap X \subset f^{-1}(B(x, \epsilon_x)) \subset f^{-1}(U).$$

Define

$$V = \bigcup_{x \in f^{-1}(U)} B(x, \delta_x)$$

which is open in $\mathbf{R}^n$, and by definition $f^{-1}(U) \subset V \cap X$. However, we have seen that $B(x, \delta_x) \cap X \subset f^{-1}(U)$ for all $x \in f^{-1}(U)$, and so

$$V \cap X = \bigcup_{x \in f^{-1}(U)} (B(x, \delta_x) \cap X) \subset f^{-1}(U).$$

We conclude that $f^{-1}(U) = V \cap X$, which concludes the proof of the theorem.

Corollary: let $f : \mathbf{R}^n \to \mathbf{R}^m$ be a function. Then f is continuous if and only if $f^{-1}(U)$ is open for all open $U \subset \mathbf{R}^m$.

$\square$

Lecture 14, 04/19/2023. This lecture is about two new properties of subsets of $\mathbf{R}^n$, and their use in the generalization of the maximum value theorem to $\mathbf{R}^n$.

Definition: 1) A subset $X \subseteq \mathbf{R}^n$ is closed if its complement $U = \mathbf{R}^n \setminus X$ is open. 2) A subset $X \subset \mathbf{R}^n$ is bounded if there exists $M > 0$ such that $||x|| \leq M$ for all $x \in X$.

Many properties of closed sets follow from properties of open sets, like in the following examples (proofs left as exercises):

1) If $\{X_i\}$ is a collection of closed subsets of $\mathbf{R}^n$, then $\bigcap_i X_i$ is closed.

2) If $\{X_1, \ldots, X_m\}$ is a finite collection of closed subsets of $\mathbf{R}^n$, then $\bigcup_{i=1}^m X_i$ is closed.

3) $\mathbf{R}^n$ is closed, and the empty set is closed.

4) Let $f : \mathbf{R}^n \to \mathbf{R}^m$ be a function. Then f is continuous is and only if $f^{-1}(X)$ is closed for all closed sets $X \subset \mathbf{R}^m$.

Closed sets admit the following alternative characterization, which is based on the notion of a limit point: If $X \subset \mathbf{R}^n$ is a subset, and $x \in \mathbf{R}^n$ is a vector, we say that x is a limit point of X if there exists a sequence $\{x_n\} \subset X$ such that $\lim_{n\to\infty} x_n = x$.

Theorem: A subset $X \subset \mathbf{R}^n$ is closed if and only if it contains all its limit points.

Proof. Assume that X is closed and $x \in \mathbf{R}^n \setminus X$. We need to prove that x is not a limit point of X. Since X is closed, we know that $\mathbf{R}^n \setminus X$ is open, and so there exists $\epsilon > 0$ such that $B(x, \epsilon) \subset \mathbf{R}^n \setminus X$. Equivalently, $B(x, \epsilon) \cap X$ is empty. If $\{x_n\}$ is a sequence in X, it follows that $x_n \notin B(x, \epsilon)$ for all n, hence $||x_n - x|| \geq \epsilon$ for all n. By definition, this implies that x_n does not converge to x. Hence x is not a limit point of X.

Conversely, assume that X is not closed. This means that there exists $x \in \mathbf{R}^n \setminus X$ such that, for all $\epsilon > 0$, the set $B(x, \epsilon) \cap X$ is not empty. Choose a point $x_n \in B(x, 1/n) \cap X$. Then $\{x_n\}$ is a sequence in X converging to x, and so x is a limit point of X not contained in X. $\square$

Example: We can use this theorem to write down many examples of closed sets. For instance, it immediately follows that sets with just one element $\{x\}$ are closed. Taking finite unions, it follows from (2) above that finite sets are closed. Taking preimages, it follows from (4) above that $f^{-1}\{x\}$ is closed if f is continuous. Hence geometric shapes like circles, lines and parabolas are closed (compare the examples in the previous lecture).

Important remark: If U is not open, it does not mean that U is closed. If U is not closed, it does not mean that U is open. For example, the interval $X = [\alpha, \beta)$ is neither open nor closed. We have already seen that X is not open. To see that it is not closed, observe that β is a limit point of X, because $\lim_{n\to\infty}(\beta - \frac{1}{n}) = \beta$ and $\beta - \frac{1}{n} \in X$ for all n large enough. Since $\beta \notin X$, this implies that X is not closed.

We closed this lecture with the following theorem, which implies the maximum value theorem for $\mathbf{R}^n$.

Theorem: Let $X \subset \mathbf{R}^n$ be closed and bounded. Let $f : X \to \mathbf{R}^m$ be continuous. Then $f(X)$ is closed and bounded.

Corollary (maximum value theorem): If $f : X \to \mathbf{R}^n$ is continuous, and X is closed and bounded, then f is bounded, and there exists $z \in X$ such that

$$||f(z)|| = \sup_{x\in X} ||f(x)||.$$

Proof. The function $\mathbf{R}^n \to \mathbf{R}$ sending x to $||f(x)||$ is continuous, since f is continuous and the norm is a continuous function $\mathbf{R}^m \to \mathbf{R}$. By the theorem above, it follows that $||f(X)||$ is closed and bounded. So there exists $M > 0$ such that $||f(x)|| \leq M$ for all $x \in X$, which means that f is bounded.

Now let $F = \sup_{x\in X} ||f(x)||$. We need to prove that there exists $z \in X$ such that $||f(z)|| = F$. By definition of the supremum, there exist $z_n \in X$ such that $||f(z_n)|| \subset (F - 1/n, F]$ for all $n > 0$. This means that $\{||f(z_n)||\}$ is a sequence in $||f(X)||$ converging to F, hence F is a limit point of $||f(X)||$. Since $||f(X)||$ is closed, this implies that $F \in ||f(X)||$. So there exists $z \in X$ such that $||f(z)|| = F$, which is what we had to prove. $\square$

Remark: Assume that $X \subset \mathbf{R}^n$ and $f : X \to \mathbf{R}^m$ is continuous. If X is closed, it does not follows that $f(X)$ is closed. For example, the function

$$\arctan : \mathbf{R} \to \mathbf{R}$$

is continuous, and $\mathbf{R}$ is closed in $\mathbf{R}$, but the image $\arctan(\mathbf{R}) = (-\pi/2, \pi/2)$ is not closed.

Similarly, if X is bounded it does not follows that $f(X)$ is bounded. For example, the function

$$\tan : (-\pi/2, \pi/2) \to \mathbf{R}$$

is continuous, and $(\pi/2, \pi/2)$ is a bounded subset of $\mathbf{R}$, but $\tan(-\pi/2, \pi/2) = \mathbf{R}$ is not bounded.

The proof of the theorem works by showing that "closed and bounded" is equivalent to another property, which is easily shown to be preserved by continuous functions.

Lecture 15, 04/21/2023. In this lecture we filled in some details from last time, and introduced compact sets.

Definition: 1) Let $X \subset \mathbf{R}^n$ be a subset. An open cover of X is a collection of open subsets $\{U_i\}$ of $\mathbf{R}^n$ such that $X \subset \bigcup_i U_i$.

2) We say that X is *compact* if every open cover $\{U_i\}$ of X has a finite subcover: this means that there exist $i_1, i_2, \dots, i_n$ such that

$$X \subset U_{i_1} \cup U_{i_2} \cup \dots \cup U_{i_n}.$$

Examples: 1) Finite sets are compact. 2) The set $X = \{1, 1/2, 1/3, 1/4, \dots\}$ is not compact. 3) The set $X_0 = X \cup \{0\}$ is compact.

The difference between X and X_0 is that in X_0 we have added to X the only limit point that wasn't in X already: this yielded a compact set. The following theorem explains why this happened.

Theorem (Heine–Borel): A subset $X \subset \mathbf{R}^n$ is compact if and only if it is closed and bounded.

Lecture 16–17, 04/24/2023 and 04/26/2023. These lectures cover the proof of the Heine–Borel theorem, which proceeds in the following three steps:

(1) Closed and bounded intervals in $\mathbf{R}$ are compact. (2) If $A \subset \mathbf{R}^n$ is compact and $B \subset \mathbf{R}^n$ is compact then the Cartesian product $A \times B \subset \mathbf{R}^{n+m}$ is compact. (3) If $X \subset \mathbf{R}^n$ is closed, $Z \subset \mathbf{R}^n$ is compact, and $X \subset Z$, then X is compact.

Putting these together implies that if X is closed and bounded then X is compact. The converse direction, i.e. the statement that if X is compact then X is closed and bounded, is given in HW5. We started with the proof of (1), which is the following theorem.

Theorem: Let $X = [\alpha, \beta] \subset \mathbf{R}$. Then X is compact.

Proof: *Calculus on manifolds*, Theorem 1-3. (Note that Spivak refers to this as the Heine–Borel theorem: we reserve that name for the more general case stated above.)

Then we assumed the following general result about compactness without proof.

Theorem: If $B \subset \mathbf{R}^n$ is compact, and K is an open cover of $\{x\} \times B$, then there exists an open set $U_x \subset \mathbf{R}^n$ containing x such that $U_x \times B$ is covered by finitely many sets in K.

Proof (optional): *Calculus on manifolds*, Theorem 1.4.

We can now give the proof of part (2), which is the following result.

Theorem: If $A \subset \mathbf{R}^n$ is compact and $B \subset \mathbf{R}^n$ is compact then the Cartesian product $A \times B \subset \mathbf{R}^{n+m}$ is compact. Proof: *Calculus on manifolds*, Corollary 1.5.

Finally, the following result completes the proof of one direction of the Heine–Borel theorem; the converse direction is covered in HW5.

Theorem: If $X \subset \mathbf{R}^n$ is closed and bounded then X is compact.

Proof. See also *Calculus on manifolds*, Corollary 1.7. Since X is bounded, there exists $M > 0$ such that $||x|| \leq M$ for all $x \in X$. By a lemma from previous lectures we deduce that

$$\max\{|x_1|, \dots, |x_n|\} \leq M$$

for all $x \in X$. This means that X is contained in a closed rectangle: more precisely,

$$X \subset [-M, M] \times \dots \times [-M, M] = Z.$$

By parts (1) and (2), every closed rectangle is compact. (See also *Calculus on manifolds*, Corollary 1.6.) Hence it suffices to prove the following general fact about compactness, which was stated as part (3) in the previous lecture: If $X \subset \mathbf{R}^n$ is closed, $Z \subset \mathbf{R}^n$ is compact, and $X \subset Z$, then X is compact.

To prove this, let $\{U_i\}$ be an open cover of X. Since X is closed, the complement $U_0 = \mathbf{R}^n \setminus X$ is open, and $\{U_i\} \cup \{U_0\}$ is an open cover of Z. Since Z is compact, there is a finite subcover: there exist indices $i_1, \dots i_n$ such that

$$Z \subset U_{i_1} \cup U_{i_2} \cup \dots \cup U_{i_n} \cup U_0.$$

Since $X \subset Z$, we also have
$$X \subset U_{i_1} \cup U_{i_2} \cup \cdots \cup U_{i_n} \cup U_0;$$
but $X \cap U_0$ is empty, by definition of U_0, and so $X \subset U_{i_1} \cup U_{i_2} \cup \cdots \cup U_{i_n}$. This is a finite subcover of $\{U_i\}$, hence X is compact. $\square$

We can now give the proof of the maximum value theorem in several variables.

Theorem: Assume that $X \subset \mathbf{R}^n$ is closed and bounded and $f : X \to \mathbf{R}^m$ is continuous. Then $f(X)$ is closed and bounded.

Proof. By the Heine–Borel theorem, X is compact, and we need to prove that $f(X)$ is compact. Let $\{U_i\}$ be an open cover of $f(X)$. Then, since f is continuous, there exist open sets V_i such that $f^{-1}(U_i) = X \cap V_i$ for all i. But we know that $f(X) \subset \bigcup_i U_i$, it follows that $X = \bigcup_i f^{-1}(U_i)$. In other words, $\{V_i\}$ is an open cover of X. So there exist $i_1, \dots, i_n$ such that
$$X \subset V_{i_1} \cup V_{i_2} \cup \cdots \cup V_{i_n}.$$
It follows from this that
$$X = (V_{i_1} \cup V_{i_2} \cup \cdots \cup V_{i_n}) \cap X = f^{-1}(U_{i_1}) \cup f^{-1}(U_{i_2}) \cup \cdots \cup f^{-1}(U_{i_n})$$
and so
$$f(X) \subset U_{i_1} \cup U_{i_2} \cup \cdots \cup U_{i_n},$$
which concludes the proof that $f(X)$ is compact. $\square$

Lecture 18, 04/28/2023. No lecture, second midterm.

Lecture 19, 05/01/2023. This lecture is about one more property of subsets of Euclidean space, and its use in the higher-dimensional generalization of the intermediate value theorem. It will be useful to recall the following characterization of intervals in $\mathbf{R}$.

Theorem: A subset $J \subset \mathbf{R}$ is an interval if and only if the following is true: for all $x, y \in J$ and all $z \in [x, y]$, we have $z \in J$.

Proof. The forward direction is immediate from the definition of an interval. For the converse, let $\alpha = \inf(J)$ and $\beta = \sup(J)$. We will prove that $J = (\alpha, \beta)$ or $[\alpha, \beta)$ or $(\alpha, \beta]$ or $[\alpha, \beta]$, depending on whether α and β are in J or not. To do this, it suffices to prove that if $\alpha < z < \beta$ then $z \in J$. But by definition of infimum and supremum there exist $x, y \in J$ such that
$$\alpha \leq x < z < y \leq \beta$$
and so $z \in J$ by our assumption on J. $\square$

Definition: A subset $X \subset \mathbf{R}^n$ is connected if whenever $U, V \subset \mathbf{R}^n$ are open sets such that

(1) $X \subset U \cup V$, and
(2) $X \cap U$ and $X \cap V$ are disjoint,

then either $X \subset U$ or $X \subset V$.

Remark: There are several ways of restating the definition. For example, X is connected if and only if for all open sets $U, V \subset \mathbf{R}^n$ such that (1) and (2) are true, we have that $X \cap U$ is empty or $X \cap V$ is empty.

Remark: If (1) and (2) hold then $X \cap U$ and $X \cap V$ are disjoint, and furthermore $X \cap U$ does not contain any limit point of $X \cap V$ (and viceversa). To see this, notice that if $\{x_n\} \subset X \cap V$ is a sequence converging to x then $x_n \notin U$ for all n, by (2). Since $\mathbf{R}^n \setminus U$ is closed, this implies that $x \notin U$. Hence x cannot be contained in $X \cap U$.

Theorem (intermediate value theorem for $\mathbf{R}^n$): Let $X \subset \mathbf{R}^n$ be connected and $f : X \to \mathbf{R}^m$ be continuous. Then $f(X)$ is connected.

Proof. Let $U, V \subset \mathbf{R}^m$ be open sets such that $f(X) \subset U \cup V$ and $f(X) \cap U$ is disjoint from $f(X) \cap V$. Then $X = f^{-1}(U) \cup f^{-1}(V)$ and $f^{-1}(U) \cap f^{-1}(V)$ is empty. Furthermore, since f is continuous, there exist open sets $W_U, W_V \subset \mathbf{R}^n$ such that
$$f^{-1}(U) = W_U \cap X, f^{-1}(V) = W_V \cap X.$$
Then $X \subset W_U \cup W_V$ and $W_U \cap X$ is disjoint from $W_V \cap X$. Since X is connected, it follows that $X \subset W_U$ or $X \subset W_V$. This implies that $f(X) \subset U$ or $f(X) \subset V$. Hence $f(X)$ is connected. $\square$

Theorem (characterization of connected subsets of $\mathbf{R}$): A subset $J \subset \mathbf{R}$ is connected if and only if J is an interval.

Proof. Assume that J is not an interval. We will prove that J is not connected. By the characterization described before, there exist $x, y \in J$ and $z \in (x, y)$ such that $z \notin J$. Let $U = (-\infty, z)$ and $V = (z, +\infty)$. Then $J \subset U \cup V = \mathbf{R} \setminus \{z\}$. However, $U \cap V$ is empty, and so certainly $(U \cap J) \cap (V \cap J)$ is empty. But since $x \notin V$ it is not true that $J \subset V$, and since $y \notin U$, it is not true that $J \subset U$; this implies that J is not connected.

The other direction is more difficult, and it is optional: see Rudin, *Principles of mathematical analysis*, Theorem 2.47 for a proof. □

Here is how to deduce the intermediate value theorem you've learned in Math 16100 from the two results above. Let $f : (\alpha, \beta) \to \mathbf{R}$ be a continuous function. By the second theorem, the interval (α, β) is connected, and by the first theorem the image $J = f(\alpha, \beta)$ is connected. By the second theorem again, J is an interval. This means that if $f(x), f(y) \in J$ and $f(x) \leq t \leq f(y)$ then $t \in f(J)$, or equivalently there exists $z \in (\alpha, \beta)$ such that $t = f(z)$. This is precisely the statement of the intermediate value theorem.

Lecture 20, 05/03/2023. Definition: Let $U \subset \mathbf{R}^n$ be an open set and let $x \in U$. A function $f : U \to \mathbf{R}^m$ is called differentiable at x if there exists a linear map $\varphi : \mathbf{R}^n \to \mathbf{R}^m$ such that

$$\lim_{h \to 0} \frac{||f(x+h) - f(x) - \varphi(h)||}{||h||} = 0.$$

Remarks: 1) The letter h denotes a vector, so you need to take the norm for the quotient to make sense. 2) We will usually work with $U = \mathbf{R}^n$, but you should know the more general definition above.

Theorem (Spivak, Theorem 2-1): If $f : \mathbf{R}^n \to \mathbf{R}^m$ and $\varphi_1, \varphi_2 : \mathbf{R}^n \to \mathbf{R}^m$ are linear functions such that

$$\lim_{h \to 0} \frac{||f(x+h) - f(h) - \varphi_i(h)||}{||h||} = 0.$$

for both $i = 1, 2$, then $\varphi_1 = \varphi_2$.

Definition: If $f : \mathbf{R}^n \to \mathbf{R}^m$ is differentiable at x, then the linear map φ is denoted $Df(x)$ and it is called the derivative of f at x. It is uniquely determined by f and x by the previous theorem. The matrix $[Df(x)]$ is called the Jacobian matrix of f at x.

We computed the following examples: (1) $f(x, y) = \sin(x)$ at (a, b). (2) $f(x, y, z) = x^2 + 2y^2 + 3z^2$ at $(1, 0, 0)$. We will see many more examples later, using two general methods for computing derivatives, namely partial derivatives and the chain rule.

Several properties of one-dimensional derivatives carry over to the higher-dimensional case, like in the following result.

Theorem: If $f : \mathbf{R}^n \to \mathbf{R}^m$ is differentiable at x, then f is continuous at x.

Proof. It suffices to prove that

$$\lim_{h \to 0} ||f(x+h) - f(x)|| = 0.$$

We know from the definition of differentiability that there exists a linear map φ such that if we define $R(h) = f(x+h) - f(x) - \varphi(h)$ then $\lim_{h \to 0} ||R(h)||/||h|| = 0$, and so $\lim_{h \to 0} ||R(h)|| = 0$. Since

$$||f(x+h) - f(x)|| \leq ||\varphi(h)|| + ||R(h)||$$

by the triangle inequality, and $\lim_{h \to 0} \varphi(h) = 0$ because φ is linear, we find that $\lim_{h \to 0} ||f(x+h) - f(x)|| = 0$. □

Lecture 21, 05/05/2023. Let $f : \mathbf{R}^n \to \mathbf{R}$ be a function, and let $x, v \in \mathbf{R}^n$. The rate of change of f at x in the direction v is defined to be

$$\lim_{t \to 0} \frac{f(x+tv) - f(x)}{t}$$

if the limit exists. Specializing this definition to $v = e_i$, we get the definition of partial derivatives.

Definition: The i-th partial derivative of f at x is

$$D_i f(x) = \frac{\partial f}{\partial x^i}(x) = \lim_{t \to 0} \frac{f(x+te_i) - f(x)}{t}$$

if the limit exists.

In other words, we fix all coordinates of x and let the i-th coordinate vary: this gives rise to a one-variable function

$$f_x^{[i]}(t) = f(x_1, \dots, x_{i-1}, t, \dots, x_n),$$

and the i-th partial derivative of f at x coincides with the derivative of $f_x^{[i]}$ at $t = x_i$.

If the i-th partial derivative exists at all $x \in \mathbf{R}^n$, then it defines a new function

$$\frac{\partial f}{\partial x^i} : \mathbf{R}^n \to \mathbf{R}.$$

This function is easy to compute if f is given by a formula, because you just have to differentiate $f_x^{[i]}$, which is the one-variable function obtained by regarding all the x_i as constants. (Everything we said so far works unchanged if the domain of f is an open set in $\mathbf{R}^n$ rather than the whole of $\mathbf{R}^n$.)

Examples: 1) $f(x, y, z) = x^2 + 2y^2 + 3z^2$. Then

$$\frac{\partial f}{\partial y}(x, y, z) = 4y \text{ and } \frac{\partial f}{\partial z}(x, y, z) = 6z.$$

2) $f(x, y) = \sin(xy^2)$. Then

$$\frac{\partial f}{\partial x}(x, y) = y^2 \cos(xy^2) \text{ and } \frac{\partial f}{\partial y}(x, y) = 2xy \cos(xy^2).$$

3) $f(x, y) = x^y$ (assuming $x > 0$). Then

$$\frac{\partial f}{\partial x}(x, y) = yx^{y-1} \text{ and } \frac{\partial f}{\partial y}(x, y) = x^y \log(x).$$

Like in the one-dimensional case, we can use partial derivatives to give necessary conditions for a point to be a local maximum or minimum.

Theorem (Spivak, Theorem 2-6): Let $f : \mathbf{R}^n \to \mathbf{R}$ be a function and $x \in \mathbf{R}^n$. Assume that $f(x) \geq f(z)$ for all z in some open set U containing x, and that $D_i f(x)$ exists. Then $D_i f(x) = 0$.

Proof. Since $D_i f(x)$ exists, the function $f_x^{[i]}$ is differentiable at x_i and its derivative at x_i equals $D_i f(x)$. Recall that $f^{[i]}(x)$ is the composition of f with the inclusion

$$i_x : \mathbf{R} \to \mathbf{R}^n, i_x(t) = (x_1, \dots, x_{i-1}, t, \dots, x_n).$$

Since i_x is continuous, the preimage $i_x^{-1}(U)$ is open, and it contains x_i. So there exists an open interval J containing x_i and contained in $i_x^{-1}(U)$. By assumption, we find that $f_x^{[i]}(x_i) \geq f(t)$ for all $t \in J$. By a result from Math 16100, this implies that the derivative of $f_x^{[i]}$ vanishes at x_i. Hence $D_i f(x) = 0$. $\square$

Warning: As in the one-dimensional case, the converse to the previous theorem can fail. However, new phenomena appear in higher dimensions, known as "saddle points". Consider the function $g : \mathbf{R}^2 \to \mathbf{R}, g(x, y) = x^2 - y^2$ at $x = 0$. Then

$$f_x^{[1]}(t) = t^2 \text{ and } f_x^{[2]}(t) = -t^2$$

so $f^{[1]}$ has a minimum at 0, but $f^{[2]}$ has a maximum at zero. Hence 0 is neither a minimum nor a maximum of f, but $D_1 f(0, 0) = D_2 f(0, 0) = 0$.

Lecture 22, 05/08/2023. The following result is the higher-dimensional version of the chain rule.

Theorem (Spivak, Theorem 2-2): Let $f : \mathbf{R}^n \to \mathbf{R}^m$ and $g : \mathbf{R}^m \to \mathbf{R}^p$ be functions. Assume that f is differentiable at α and g is differentiable at $f(\alpha)$. Then $g \circ f$ is differentiable at α, and

$$D(g \circ f)(\alpha) = Dg(f(\alpha)) \circ Df(\alpha) : \mathbf{R}^n \to \mathbf{R}^p.$$

Proof. Abbreviate notation by $\lambda = Df(\alpha), \mu = Dg(f(\alpha))$ and define three remainder terms by

(1) $\phi(x) = f(x) - f(\alpha) - \lambda(x - \alpha)$,
(2) $\psi(y) = g(y) - g(f(\alpha)) - \mu(y - f(\alpha))$,
(3) $\rho(x) = gf(x) - gf(\alpha) - \mu\lambda(x - \alpha)$.

The definition of the derivative implies that

$$\lim_{x\to\alpha}\frac{||\phi(x)||}{||x-\alpha||}=0,\ \text{and}\ \lim_{y\to f(\alpha)}\frac{||\psi(y)||}{||y-f(\alpha)||}=0.$$

We need to prove that

$$\lim_{x\to\alpha}\frac{||\rho(x)||}{||x-\alpha||}=0.$$

We begin rewriting $\rho(x)$: we have

$$\rho(x)=gf(x)-gf(\alpha)-\mu\lambda(x-\alpha)=gf(x)-gf(\alpha)-\mu(f(x)-f(\alpha)-\phi(x))$$

by definition of ϕ. Using the linearity of μ we deduce

$$\rho(x)=g(f(x))-g(f(\alpha))-\mu(f(x)-f(\alpha))+\mu(\phi(x))$$

and by definition of ψ we find

$$\rho(x)=\psi(f(x))+\mu(\phi(x)).$$

Recall that since μ and λ are linear there exists $M>0$ such that $||\mu(z)||\le M||z||$ and $||\lambda(t)||\le M||t||$ for all z,t. Together with the remainder property for ϕ, this implies that

$$\lim_{x\to\alpha}\frac{||\mu(\phi(x))||}{||x-\alpha||}=0.$$

The other term is more complicated. Fix $\epsilon>0$. The definition of ψ implies that for some $\delta>0$ we have

$$||f(x)-f(\alpha)||<\delta \text{ implies } ||\psi(f(x))||<\epsilon||f(x)-f(\alpha)||.$$

But we have proved last time that f is continuous at α, since it is differentiable at α. Hence there exists $\delta_1>0$ such that $||x-\alpha||<\delta_1$ implies $||f(x)-f(\alpha)||<\delta$, and so $||x-\alpha||<\delta_1$ implies that

$$||\psi(f(x))||<\epsilon||f(x)-f(\alpha)||=\epsilon||\phi(x)+\lambda(x-\alpha)||\le\epsilon||\phi(x)||+\epsilon M||x-\alpha||.$$

Dividing through by $||x-\alpha||$ and letting $||x-\alpha||\to 0$ we find that

$$\frac{||\psi(f(x))||}{||x-\alpha||}\le 2\epsilon M$$

whenever $||x-\alpha||$ is small enough. Since $\epsilon>0$ was arbitrary, this implies that

$$\lim_{x\to\alpha}\frac{||\psi(f(x))||}{||x-\alpha||}=0.$$

□

Next we are going to show how the chain rule can be used to compute derivatives in terms of partial derivatives. We begin with an intermediate result, reducing the problem to the case of functions valued in $\mathbf{R}$.

Theorem (Spivak, Theorem 2-3(3)): A function $f:\mathbf{R}^n\to\mathbf{R}^m$ is differentiable at $\alpha\in\mathbf{R}^n$ if and only if the coordinate $f_i:\mathbf{R}^n\to\mathbf{R}$ is differentiable for all $1\le i\le m$. If this is true, then the i-th row of $[Df(\alpha)]$ is $[Df_i(\alpha)]$.

Proof. One direction is immediate by the chain rule. By HW7 the coordinate functions $c_i:\mathbf{R}^m\to\mathbf{R}$ are differentiable everywhere, because they are linear, and so the composite function $f_i=c_i\circ f$ is differentiable at α if f is differentiable at α.

Conversely, assume that f_i is differentiable at α for all $1\le i\le m$. Let $\lambda:\mathbf{R}^n\to\mathbf{R}^m$ be the linear map defined by

$$\lambda(x)=(Df_1(\alpha)(x),Df_2(\alpha)(x),\ldots,Df_m(\alpha)(x))$$

so that the i-th row of $[\lambda]$ is $[Df_i(\alpha)]$. We will prove from the definition that $Df(\alpha)=\lambda$. Namely, since

$$||f(\alpha+h)-f(\alpha)-\lambda(h)||\le\sqrt{m}\max_{1\le i\le m}|f_i(\alpha+h)-f_i(\alpha)-Df_i(\alpha)(h)|$$

and

$$\lim_{h\to 0}\frac{|f_i(\alpha+h)-f_i(\alpha)-Df_i(\alpha)(h)|}{||h||}=0$$

for all i by definition of Df_i, we deduce that

$$\lim_{h \to 0} \frac{||f(\alpha+h) - f(\alpha) - \lambda(h)||}{||h||} = 0.$$

This means that f is differentiable at α and that $Df(\alpha) = \lambda$, which implies the claim. □

Lecture 23, 05/10/2023. This lecture explains how to use partial derivatives and the chain rule to compute matrix derivatives.

Theorem (Spivak, Theorem 2-7): If f is differentiable at α then $\partial f_i/\partial x^j(\alpha)$ exists for all i, j, and it coincides with the ij-th entry of the Jacobian matrix $[Df(\alpha)]$.

Proof. Recall that $\partial f_i/\partial x^j(\alpha)$ is the derivative at α_j of the composite function $f_i \circ i_\alpha^{[j]}$, where

$$i_\alpha^{[j]} : \mathbf{R} \to \mathbf{R}^n, i_\alpha^{[j]}(t) = (\alpha_1, \ldots, \alpha_{j-1}, t, \alpha_{j+1}, \ldots, \alpha_n).$$

This function $i_\alpha^{[j]}$ is differentiable by the theorem above and HW7, since its coordinates are either constant or linear, and its Jacobian at $t = \alpha_j$ is the column vector e_j. Similarly, the function f_i is differentiable at $\alpha = i_\alpha^{[j]}(\alpha_j)$. By the chain rule we find that $f_i \circ i_\alpha^{[j]}$ is differentiable at α_j, and its derivative is the product

$$[Df_i(\alpha)](e_j).$$

This is the j-th entry of the row vector $[Df_i(\alpha)]$, which by the previous theorem coincides with the i-th row of the matrix $[Df(\alpha)]$. Hence $\partial f_i/\partial x^j(\alpha)$ is the ij-th entry of $[Df(\alpha)]$. □

Remark: The converse to this theorem is not true. For example, the function defined by

$$f(x, y) = \frac{xy}{x^2 + y^2} \text{ if } (x, y) \neq (0, 0)$$

and $f(0, 0) = 0$ has both partial derivatives everywhere on $\mathbf{R}^2 \setminus 0$, by the quotient rule. Furthermore, both partial derivatives exist at $(0, 0)$ by an explicit computation done in class. If the converse to the theorem above were true, it would follow that f is differentiable at $(0, 0)$. However, it is not even continuous at $(0, 0)$, for example because

$$\lim_{n \to \infty} f(1/n, 1/n) = 1/2 \neq f(0, 0).$$

However, the following result provides a partial converse which is often enough in applications.

Theorem (Spivak, Theorem 2-8): Let $f : \mathbf{R}^n \to \mathbf{R}^m$, choose $\alpha \in \mathbf{R}^n$, and let $f_i = c_i \circ f$ be the i-th coordinate of f. If for all i, j the partial derivative $\partial f_i/\partial x^j(z)$ exists for all z in some open set U containing α, and the function $\partial f_i/\partial x^j(z)$ is continuous at α, then f is differentiable at α. (Such a function is said to be *continuously differentiable* at α.)

Lecture 24, 05/12/2023. In this lecture we proved the previous theorem and then gave a few examples of partial derivatives and the chain rule.

1) We gave two proofs that polynomial functions are differentiable. The first one is longer and uses the following general result. The second is a direct application of partial derivatives.

Theorem (Spivak, Theorem 2-3): Let $f, g : \mathbf{R}^n \to \mathbf{R}$ be differentiable functions at $\alpha \in \mathbf{R}$. Then $f + g, fg$ and λg are differentiable at α, for all $\lambda \in \mathbf{R}$, and their derivatives are given by

$$D(f+g)(\alpha) = Df(\alpha) + Dg(\alpha)$$
$$D(fg)(\alpha) = g(\alpha)Df(\alpha) + f(\alpha)Dg(\alpha)$$
$$D(\lambda f)(\alpha) = \lambda Df(\alpha).$$

Proof. Define two functions

$$\text{sum} : \mathbf{R}^2 \to \mathbf{R}, \text{sum}(x, y) = x + y$$

$$\text{prod} : \mathbf{R}^2 \to \mathbf{R}, \text{prod}(x, y) = xy.$$

The functions f and g can be assembled into a function

$$(f, g) : \mathbf{R}^n \to \mathbf{R}^2, (f, g)(x) = (f(x), g(x))$$

and by definition we have $f + g = \mathrm{sum} \circ (f, g)$ and $fg = \mathrm{prod} \circ (f, g)$. The function sum is differentiable, because it is linear, and its Jacobian matrix at $(x, y) \in \mathbf{R}^2$ is independent of (x, y) and equal to $\begin{pmatrix} 1 & 1 \end{pmatrix}$. The function prod is not linear, but using partial derivatives we see that it is differentiable everywhere and

$$[D\,\mathrm{prod}(x, y)] = \begin{pmatrix} y & x \end{pmatrix}.$$

Since

$$[D(f, g)(\alpha)] = \begin{pmatrix} [Df(\alpha)] \\ [Dg(\alpha)] \end{pmatrix}$$

we see that the first two parts of the theorem follow by the chain rule. The third part is left as an exercise. □

2) We saw how to compute the Jacobian matrices of elementary functions using partial derivatives and the theorem from last time. For example,

$$f(x, y) = \left(\frac{y}{1 + x^2}, \sin(xy) \right)$$

is differentiable and its Jacobian at (x, y) is given by

$$\begin{pmatrix} \frac{-2xy}{(1+x^2)^2} & \frac{1}{1+x^2} \\ y\cos(xy) & x\cos(xy). \end{pmatrix}$$

Lecture 25, 05/15/2023. Chain rule for partial derivatives: see Spivak, Theorem 2-9.

Example: if $f : \mathbf{R}^2 \to \mathbf{R}$ is a differentiable function, then the partial derivatives of

$$F(x, y, z) = f(x^2 - y + 2yz^2, z^3 e^{xy})$$

can be computed in the following way. Let $g(x, y, z) = (x^2 - y + 2yz^2, z^3 e^{xy})$, so that $F = f \circ g$. Then compute the Jacobian of g at $\alpha = (x, y, z)$. It is equal to

$$[Dg(\alpha)] = \begin{pmatrix} 2x & 2z^2 - 1 & 4yz \\ yz^3 e^{xy} & xz^3 e^{xy} & 3z^2 e^{xy}. \end{pmatrix}$$

The chain rule implies that $\partial F/\partial x(\alpha)$ is the first entry in the matrix product

$$\begin{pmatrix} \frac{\partial f}{\partial x}(g(\alpha)) & \frac{\partial f}{\partial y}(g(\alpha)) \end{pmatrix} [Dg(\alpha)].$$

Doing the computation, we find that

$$\frac{\partial F}{\partial x}(\alpha) = \frac{\partial f}{\partial x}(g(\alpha))2x + \frac{\partial f}{\partial y}(g(\alpha)) yz^3 e^{xy}.$$

The examinable material for the final ends here.

We ended our discussion of multivariable differentiation with the definition of higher-order and mixed partial derivatives, the statement of the theorem on equality of mixed partials (Spivak, Theorem 2-5), and the definition of smooth functions. We also gave the statement of the inverse function theorem (Theorem 2.11).

Lecture 26, 05/17/2023. Multivariable integration, not examinable. (Definitions, ϵ-criterion for integrability, and statement of Fubini's theorem.)

Lecture 27, 05/19/2023. Revision class.